116 Ethical Hacking Essential Concepts - I

EC-Council CEH v13

Internet Control Message Protocol (ICMP)

- IP is an unreliable method for the delivery of network data
- It does not notify the sender of **failed data transmission**
- Internet Control Message Protocol (ICMP) is the component of the TCP/IP protocol stack that addresses this basic limitation of IP
- ICMP does not overcome the **unreliability issues in IP**
- Reliability, if required, must be provided by upper-layer protocols (TCP or the application)

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Error Reporting and Correction

When datagram delivery errors occur, **ICMP reports** the following errors back to the source of the datagram:

Workstation 1 sends a datagram to Workstation 6

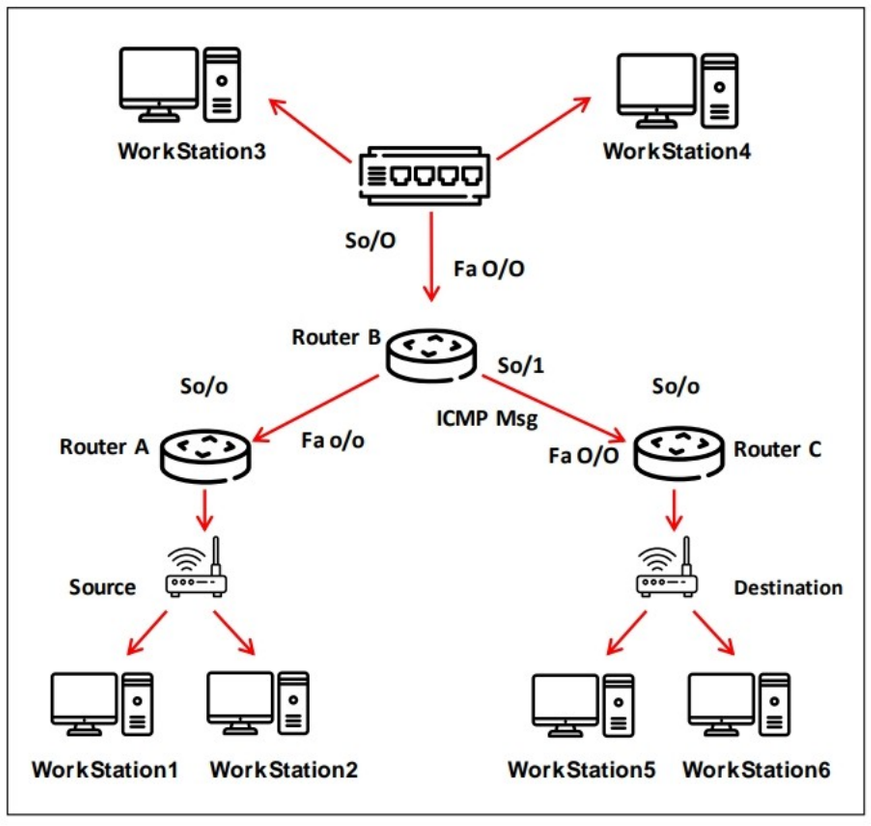
Fa0/0 on Router C goes down

Router C then utilizes ICMP to send a message indicating that the datagram could not be delivered back to Workstation 1

ICMP does not correct the encountered network problem

Router C knows only the source and destination IP addresses of the datagram

ICMP reports on the status of the delivered packet only to the source device



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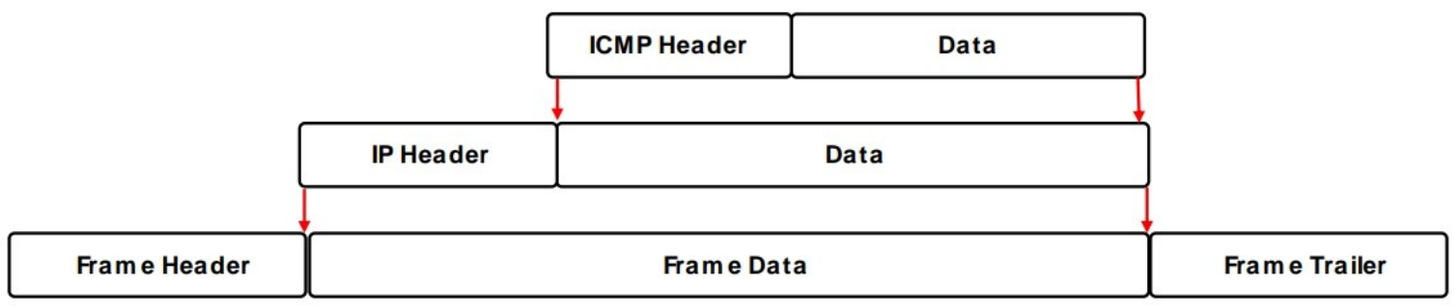
ICMP Message Delivery

- ICMP messages are encapsulated into the **datagram**
- Encapsulation uses the same technique IP uses to **deliver data**, which is subject to the same delivery failures as any IP packet
- This creates a scenario where error reports could generate more error reports
- This causes increased congestion within an **already ailing network**
- Errors created by ICMP messages do not generate their **own ICMP messages**
- It is possible to have a datagram delivery error that is never reported back to the **sender of the data**



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Format of an ICMP Message

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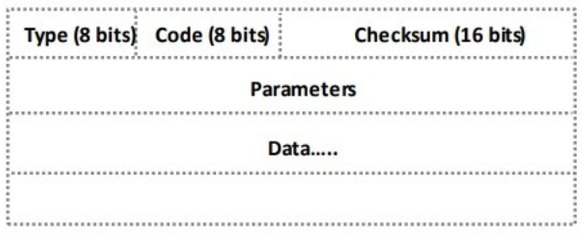
Type	Name
0	Echo Reply
1	Unassigned
2	Unassigned
3	Destination Unreachable
4	Source Quench
5	Redirect
6	Alternate Host Address
7	Unassigned
8	Echo
9	Router Advertisement
10	Router Solicitation
11	Time Exceeded
12	Parameter Problem
13	Timestamp
14	Timestamp Reply
15	Information Request
16	Information Reply
17	Address Mask Request
18	Address Mask Reply
19	Reserved (for Security)
20-29	Reserved (for Robustness Experiment)
30	Traceroute
31	Datagram Conversion Error
32	Mobile Host Redirect
33	IPv6 Where-Are-You
34	IPv6 I-Am-Here
35	Mobile Registration Request
36	Mobile Registration Reply
37	Domain Name Request
38	Domain Name Reply
39	SKIP
40	Photuris
41-255	Reserved

Code Field

Type 3: Destination Unreachable

Codes

0 Net Unreachable
1 Host Unreachable
2 Protocol Unreachable
3 Port Unreachable
4 Fragmentation Needed and Don't Fragment was Set
5 Source Route Failed
6 Destination Network Unknown
7 Destination Host Unknown
8 Source Host Isolated
9 Communication with Destination Network is Administratively Prohibited
10 Communication with Destination Host is Administratively Prohibited
11 Destination Network Unreachable for Type of Service
12 Destination Host Unreachable for Type of Service
13 Communication Administratively Prohibited
14 Host Precedence Violation
15 Precedence cutoff in effect



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